📊 Power BI Project – E-Commerce Sales & Customer Analysis Dashboard

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📌 Project Overview

This project is a **5-page interactive Power BI dashboard** built on the same E-Commerce dataset used in my SQL analysis project. The goal was to go beyond SQL queries and turn raw data into **visual, business-ready insights** that a non-technical stakeholder could immediately understand and act on.

> 💡 This dashboard complements the SQL E-Commerce Analysis project — the SQL project extracts insights through queries; this dashboard presents them visually with interactive filters.

🗂️ Dataset

Field	Detail
Records	50,000+ transactions
Key Metrics	Total Revenue: 339M | Total Customers: 25K | Total Orders: 124K | Avg Order Value: 2.72K
Categories	Books, Clothing, Electronics, Home
Payment Methods	Credit Card, PayPal, Cash, Crypto

📄 Dashboard Pages

Page 1 — Executive KPI Overview

Business Question: What is the overall health of the business?

- KPI cards: Total Revenue (339M), Total Customers (25K), Total Orders (124K), Average Order Value (2.72K)
- Revenue trend line by month — identifies seasonal peaks and drops
- Revenue by Product Category (bar chart)
- Revenue by Payment Method (donut chart) — Credit Card leads at 40.19%
- **Interactive Gender filter** — slice all visuals by Male/Female

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Page 2 — Product Category Deep Dive

Business Question: Which product categories are driving revenue, volume, and returns?

- Revenue by Product Category — Books and Clothing lead
- Quantity sold by Product Category — Clothing highest in volume
- Average Order Value by Category — consistent ~2.7K across all categories
- Returns by Product Category — Books and Clothing also have highest returns
- **Interactive Product Category slicer**

Page 3 — Customer Segmentation

Business Question: Who are our most valuable customers and how do demographics affect spending?

- Revenue by Gender — nearly equal split (Female 50.26% vs Male 49.74%)
- Revenue by Age Group — 46+ segment contributes the most revenue (~0.3bn)
- Total Revenue by Age Group and Gender — clustered bar for cross-analysis
- Top 10 customers by revenue table — John Smith leads at 286,353
- Revenue by Payment Method — Credit Card dominates at 40.24%

Page 4 — Churn Analysis

Business Question: How much revenue are we losing to churned customers and which segments churn most?

- Active vs Churned customer count — ~40K active vs ~10K churned
- Revenue by Churn Status — active customers generate ~5x more revenue
- Churn by Product Category — Clothing and Books have highest churn counts
- Churn by Age Group — 46+ segment has the most churned customers in absolute numbers
- **Actionable Insight:** Retention campaigns should focus on Clothing/Books buyers in the 46+ segment

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Page 5 — Returns Analysis

Business Question: Where are returns concentrated and what drives return behaviour?

- Total Returns: 101K
- Returns by Age Group — evenly distributed across all age segments
- Returns by Payment Method — uniform across Credit Card, PayPal, Cash, Crypto
- Returns by Product Category — Books and Clothing lead in return volume

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🧠 Key Business Insights

#	Insight	Business Recommendation
1	Credit Card is used in 40%+ of all transactions	Offer Credit Card-exclusive discounts to increase AOV
2	46+ age group generates the most revenue	Target premium products and loyalty programs at this segment
3	Churned customers = ~20% of customer base	Launch re-engagement email campaigns for churned segment
4	Books & Clothing have highest returns	Investigate product quality/description accuracy for these categories
5	Revenue split is almost equal between genders	Marketing campaigns don't need heavy gender segmentation
6	Revenue dips sharply after June	Plan promotional campaigns for Q3 to counter the mid-year slump

🛠️ Tools & Features Used

Feature	Used
KPI Cards	✅
Line Charts (Trend)	✅
Bar & Clustered Bar Charts	✅
Donut / Pie Charts	✅
Interactive Slicers (Gender, Category)	✅
Data Tables	✅
Multi-page Report	✅ (5 pages)
Cross-visual Filtering	✅

📁 File Structure

```
powerbi-ecommerce-dashboard/
│
├── README.md
├── dashboard_screenshots/
│   ├── page1_kpi_overview.png
│   ├── page2_product_category.png
│   ├── page3_customer_segmentation.png
│   ├── page4_churn_analysis.png
│   └── page5_returns_analysis.png
└── power_bi_project.pdf        ← Full dashboard export
```

🔗 Related Project

This dashboard is built on the same dataset as my SQL project: 👉 SQL E-Commerce Customer Data Analysis

Together they demonstrate the **full analytics workflow**: SQL for data extraction and insight generation → Power BI for visual storytelling and business reporting.

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